

Class Activity

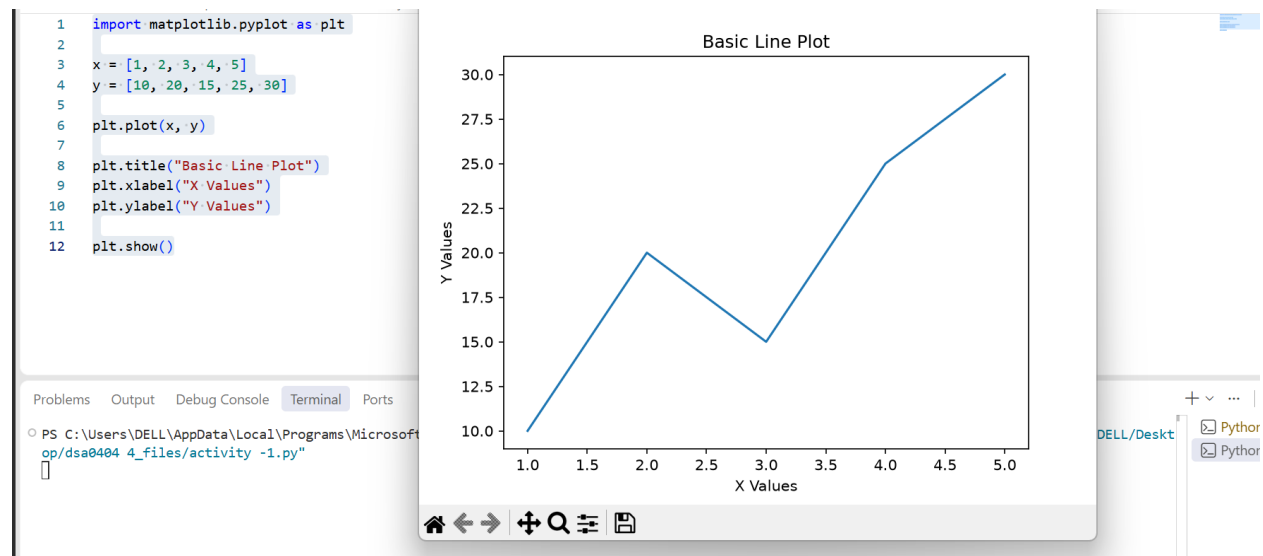
Name: Sania Herbert

Reg No:192324062

Submitted to: Dr. B.Shyamala Gowri

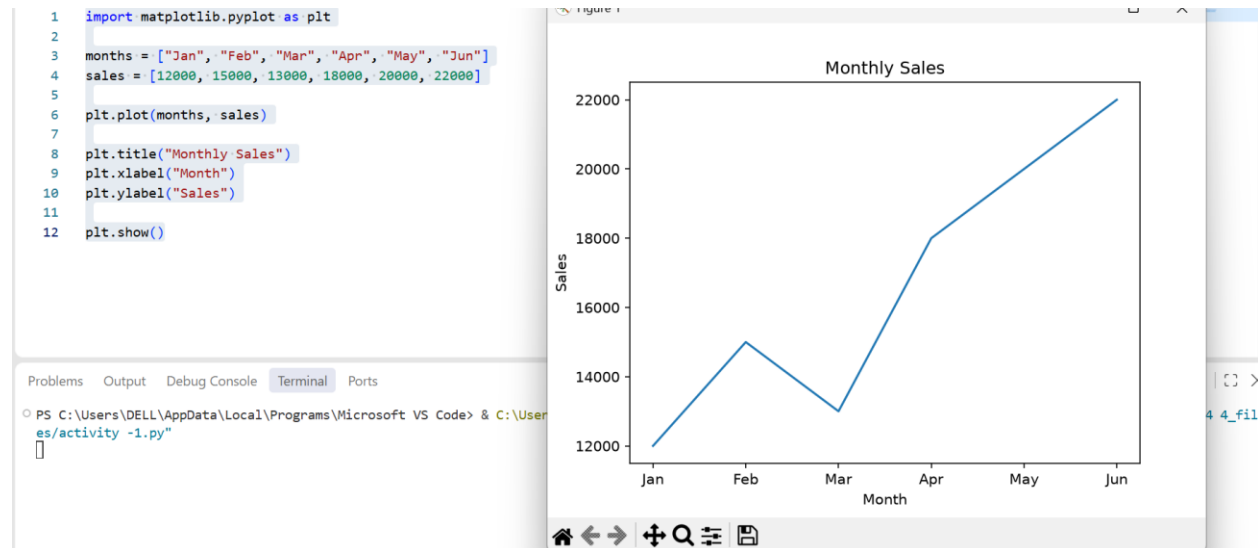
Que1:

Write a Python program using Matplotlib to create a basic line plot for the given x and y values.



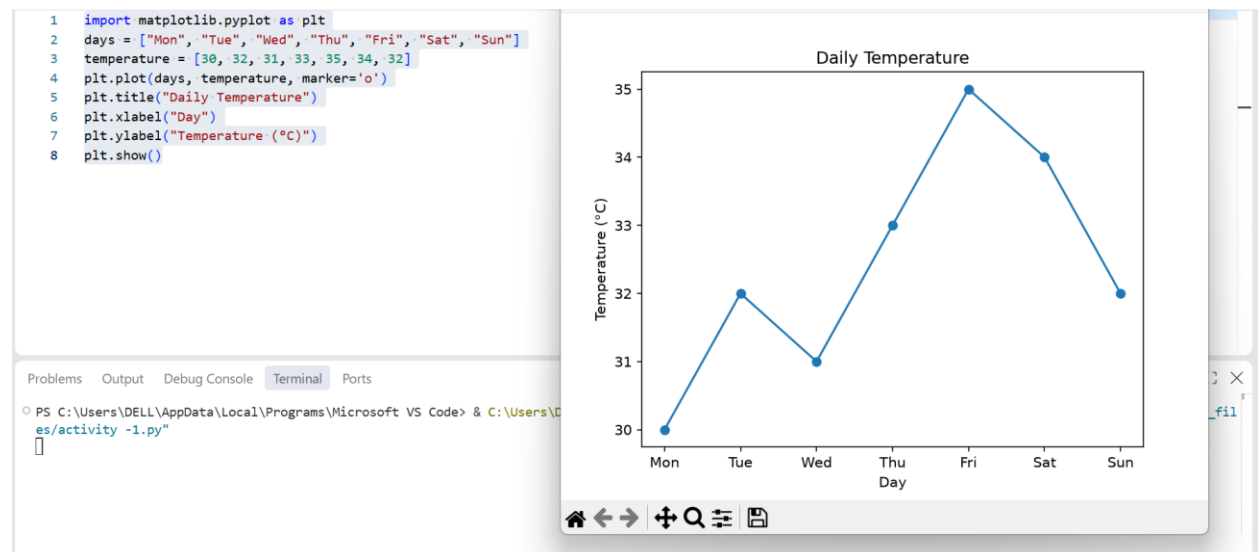
Que:2

Write a Python program to plot monthly sales data and add a suitable title, x-axis label, and y-axis label.



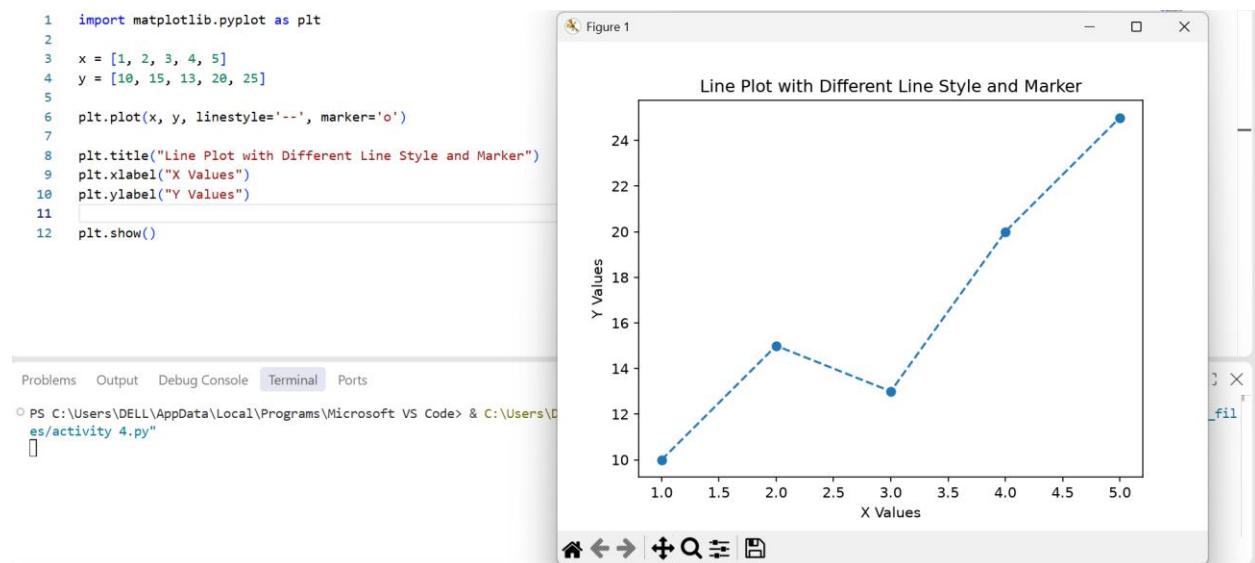
Que 3:

Write a Python program to display daily temperature values using a line plot with suitable markers.



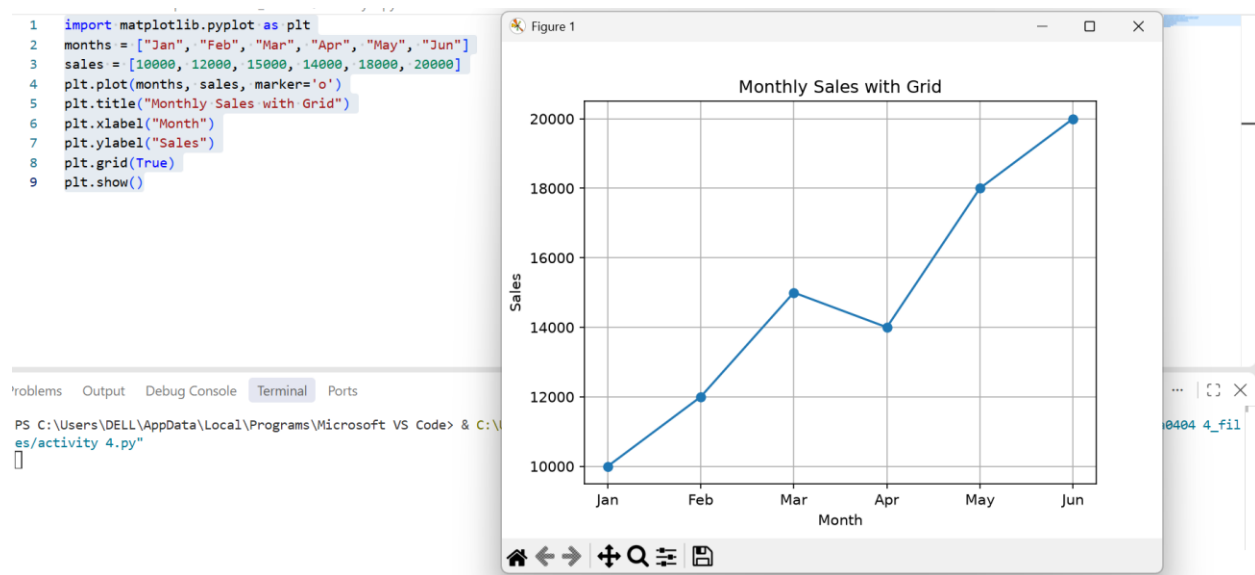
Que 4:

Write a Python program to create a line plot using different line styles and markers and display an appropriate title.



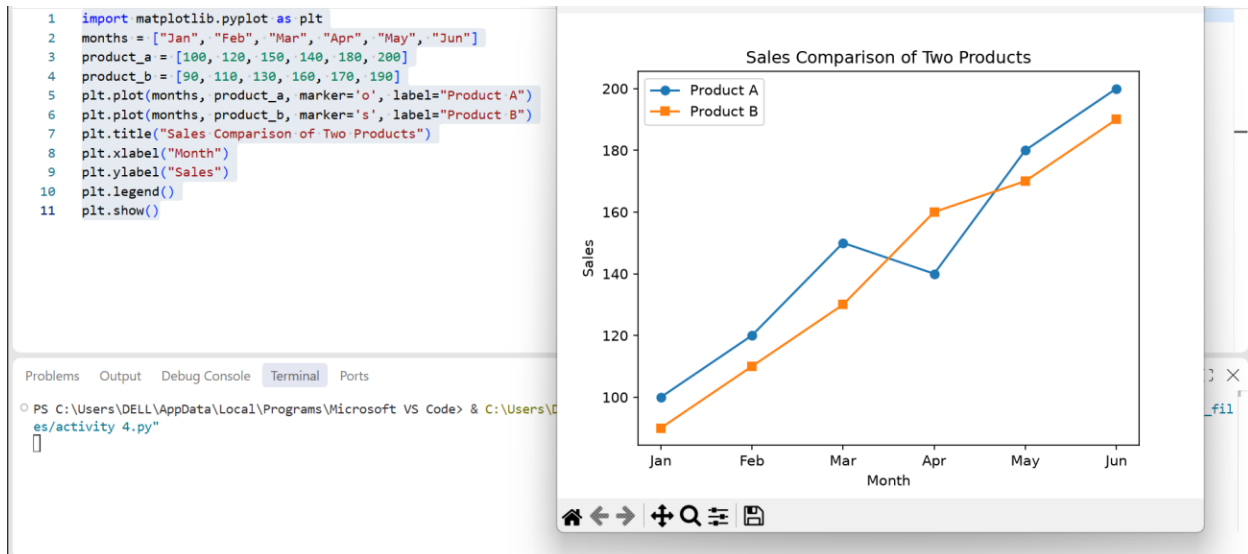
Que 5:

Write a Python program to visualize monthly sales using a line plot and add grid lines to improve readability.



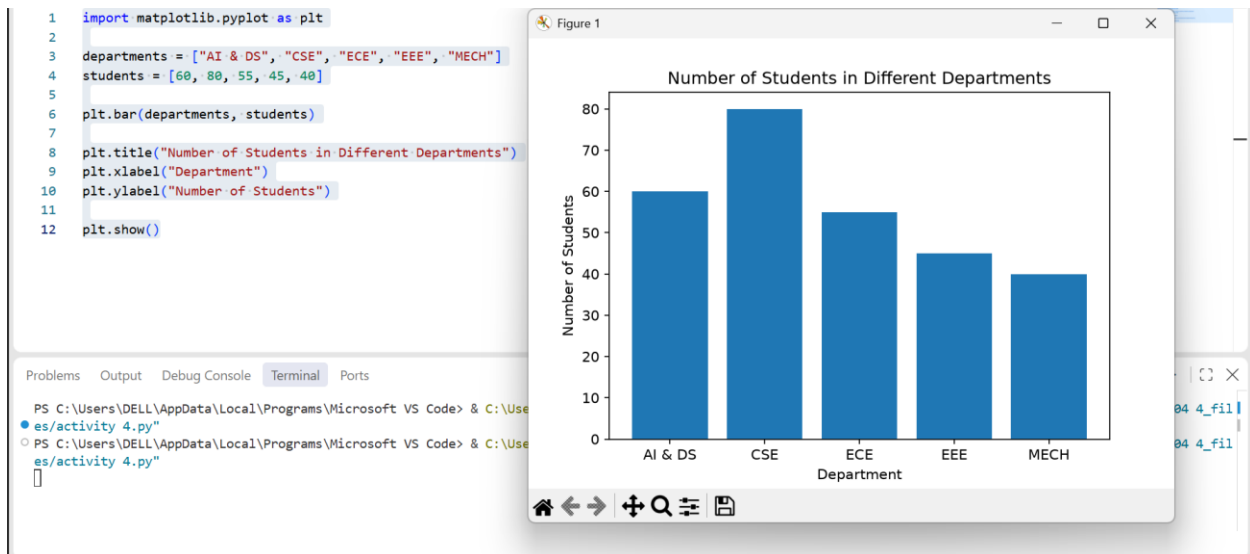
Que :6

Write a Python program to compare the sales of two products using two lines on the same graph. Add appropriate labels and a legend.



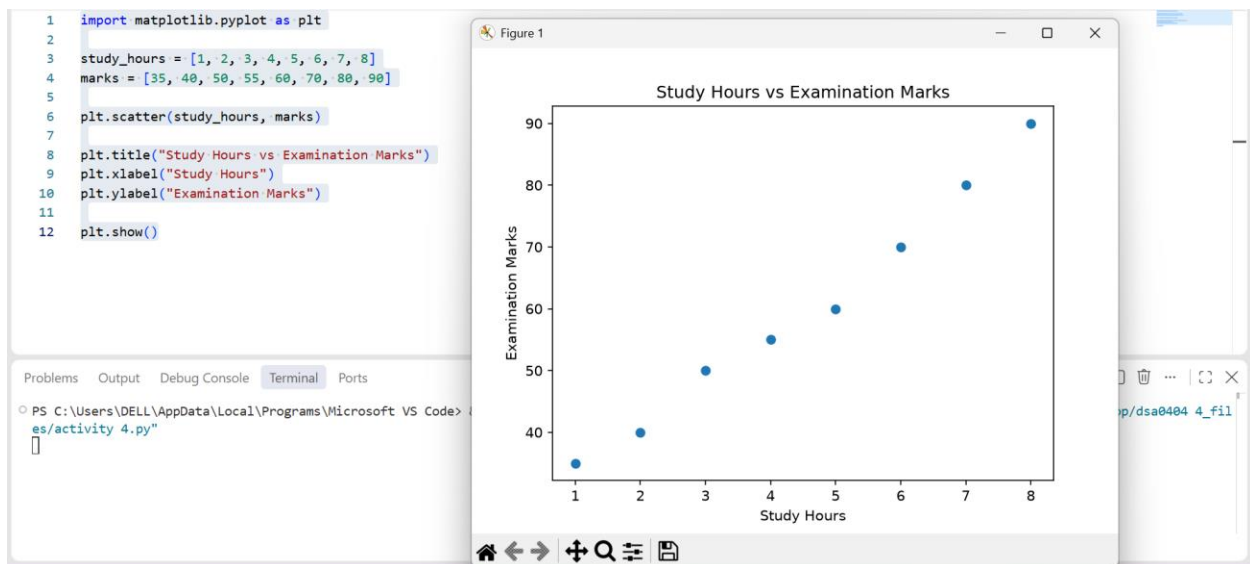
Que:7

Write a Python program to create a bar chart showing the number of students in different departments. Add a title and appropriate axis labels.



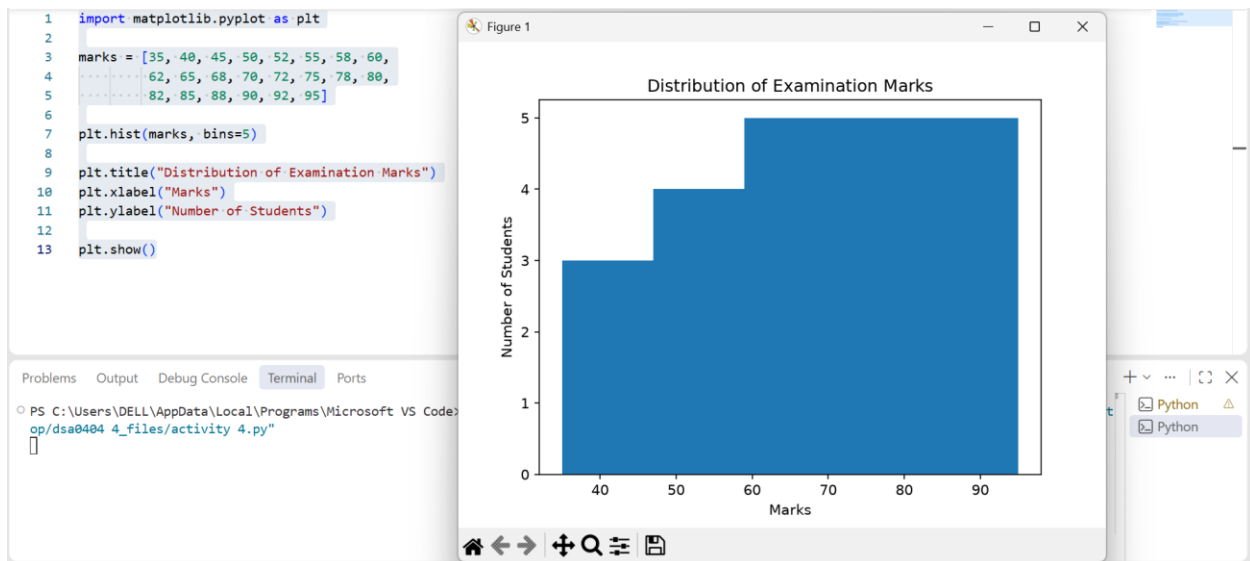
Que: 8

Write a Python program to create a scatter plot showing the relationship between students study hours and examination marks.



Que: 9

Write a Python program to create a histogram representing the distribution of students examination marks.



Que: 10 Write a Python program to create a complete Matplotlib visualization comparing monthly sales for two years, using a suitable figure size, markers, line styles, title, axis labels, legend, and grid.

```
1 import matplotlib.pyplot as plt
2 months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
3           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]
4 sales_2025 = [12000, 14000, 15000, 16000, 18000, 20000,
5              19000, 21000, 22000, 23000, 25000, 27000]
6 sales_2026 = [13000, 15000, 16000, 17000, 19000, 21000,
7              22000, 23000, 24000, 25000, 27000, 29000]
8 plt.figure(figsize=(10, 6))
9 plt.plot(
10     months,
11     sales_2025,
12     marker='o',
13     linestyle='-',
14     label="2025"
15 )
16 plt.plot(
17     months,
18     sales_2026,
19     marker='s',
20     linestyle='--',
21     label="2026"
22 )
23 plt.legend()
24 plt.show()
```

Problems Output Debug Console Terminal Ports

PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> & C:\Users\DELL\Desktop\dsa0404_4_files\activity 4.py"

● ELL\Desktop\dsa0404_4_files\activity 4.py"

○ PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> & C:\Users\DELL\Desktop\dsa0404_4_files\activity 4.py"

□

